Applied Science Department

The Walschaerts Valve Gear.

III.—Construction.

In view of the fact that the Walschaerts valve gearing does not possess that flexibility of adjustment common to shifting link valve gearings, there is greater need for perfect accuracy in construction. Not only must the design be carefully laid out, but in fitting up the gearing a degree of exactness must be attained that approaches as nearly as possible to perfection. As a general rule it is very safe to presume that the design, as far as the draftsman's work is concerned, is correct, but the same cannot always be said of the machinists' work. Shopmen are well aware that in the process of hardening the wearing parts of machines there is a tendency to irregularities, owing to the variations in the sizes of the parts. This is particularly true in regard to the parts where the greatest degree of exactness is required, and it should be carefully observed that the radial link attachment, extending as it does some distance beyond the link proper, has not moved in one direction or the other during the hardening process. The application of a turnbuckle or other device to the eccentric rod, although not now used, was a ready corrective to variations of this kind, but it was not a complete remedy, as it will be readily understood that any variation in the paths described by the moving parts have some disturbing effect on the constructor's design.

Errors of this kind are more easily detected than remedied, and it may be accepted as a rule that organic defects in the construction of the Walschaerts valve gearing can rarely be altogether remedied. It is a well-known fact among shop men that in the construction of locomotives there is a tendency to what may properly be called dramatic flourishes in the beginning of the building of a locomotive. The frames and saddle and cylinders are bolted together with a degree of rapidity that fills the eye that is pleased with superficial appearances. The exact alignment of the parts, however, is never correct by chance. Whatever is correct in mechanism is always difficult of accomplishment, and it is time well spent to note that the pedestal jaws of the frames are not only exactly square to each other, but that the shoes are also perfectly parallel. It should not be supposed that the simple matter of adding or subtracting a certain quantity of metal to or from the shoes will restore or maintain a just relation of the parts to each other. The least variation in the setting of the frames has the pernicious effect of throwing the bolt holes in the frames out of position, and braces and guide yokes and rocker boxes are more or less strangers to the truth from the first day of their industrial existence, and no amount of temporary tinkering can ever bring them back to where they properly belong. This is especially true of the relation of the center of the main driving axles to the various parts of the Walschaerts valve gearing, and it cannot be too strongly instilled into the minds of those who are engaged in this important work that every precaution should be taken to maintain the exact mathematical relation of the parts to which the valve gearing is attached and from which it derives its accuracy of movement. This is the highest kind of constructive work, and to which the work of the draughtsman is merely preliminary. It is a singular circumstance that error multiplies as it proceeds from point to point, and that which is only a small fraction at the main axle becomes a considerable quantity by the time that it has reached the valve opening.

In the designing of the Walschaerts valve gearing the length of the piston stroke being given, together with the amount of valve travel and the extent to which the valve overlaps the steam ports, and also the amount of opening or lead which may be deemed necessary, the combination lever is designed in such proportion as to move the valve the exact amount of lap and lead away from the central position at the time that the piston is at the extreme end of the stroke. The union link and crosshead arm are so constructed that the combination lever will be in the perpendicular position when the piston is in the center of the cylinder. The radial link is so constructed that its arc or curve corresponds to the circle described by the length of the radius bar; that is, measuring from the point to where the radius bar is attached to the combination lever along to the center of the link block. The radius bar should be at least eight times the length of the space in which the link block is designed to travel. The longer the radius bar is the more direct its thrust will be on the valve rod, and consequently less subject to the disturbing and distorting effect of a short angular movement. The length of the link is such as when oscillated by the eccentric rod the radius bar will move the slide valve the desired amount of travel; that is, when the link block which is attached to the radius bar and moves freely in the link is at either end of the link.

As we have already stated in the case of engines equipped with the ordinary slide valve or with outside admission piston valves, the radius bar is connected to the combination lever below the valve stem, and in the case of piston valves where the steam is admitted from the inner edge of the valve, called inside admission, the connection of the radius bar is above the valve stem. It should be noted that in order to maintain the perfect equality of the valve travel both in the forward and backward motions the suspension point of the link should be in line with a point between the combination lever's connection with the radius bar and valve rod, the line being drawn parallel with the valve rod. The same

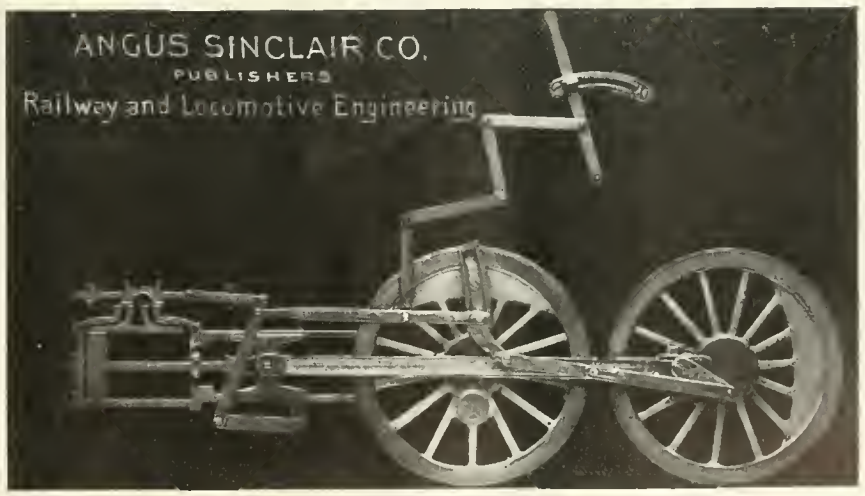


FIG. 1. FORWARD GEAR, LINK-BLOCK AT BOTTOM OF LINK.

alignment should be maintained as nearly as possible in regard to the eccentric rod and the center of the driving axle. The point of connection of the eccentric rod with the link should be parallel with the axle centers. In the construction of some classes of locomotives, especially with the largest kind of cylinders, the perfect alignment of these points would necessitate an extra extension of the link arm, which in turn would require an excessive amount of eccentric throw, and hence the point of eccentric rod connection with the link arm is often at some distance above the centre line. The distorting influence of these variations is reduced to a minimum by the action of the combination lever, which being attached to the crosshead, maintains the position of the valve with a degree of accuracy rarely equalled by any kind of mechanism where circular motion is changed to linear or reciprocating motion.

A working model of the Walschaerts valve gearing is a good aid to a clear understanding of the mechanism, and readily reveals not only its essential features and advantages, but shows at a glance some of the seeming contradictions to other valve gearings that its actions exhibit. Thus, in Fig. 1 an illustration of the model before us, it will be noted that in the forward motion the link block is in the bottom of the link. This is opposite to what occurs in the Stephenson shifting link, but it will be observed at once that it is the link block and attached radius rod that moves, and the action is direct and simple in comparison with a floating link passing through a variable arc, disturbing the exact position of the valve as it passes from point to point. It may be stated here that in Walschaerts' valve gear in the position shown in Fig. 1 the reverse lever may be moved from one end of the sector or quadrant to the other end without disturbing the position of the valve. If the eccentric rod is of the proper length it will be readily proved at either end of the piston stroke by moving the reverse lever and thereby moving the link block in the link. If the valve remains stationary the rod referred to is of the correct length. If the piston is at the extreme front end of the cylinder and the valve moves forward as the link block rises in the link the eccentric rod is too short. If on the other hand the valve moves backward the eccentric rod is too long. The eccentric rod controls the position and motion of the link, and should be so adjusted in point of length that the link block and accompanying radius bar should not disturb the valve when the piston is at the extreme ends of the stroke.

A peculiarity of the gearing may be noted at this point which affects the movement of the valve advantageously. As shown in Fig. 1, the eccentric rod is on the bottom center, and when in motion moves the link at a comparatively high rate of speed, the motion of the link being reduced to a minimum as the eccentric rod passes around what are properly called the dead centers in its circular path. It will thus be seen that at the moment when the eccentric or return crank is at the lowest or highest point of its path the valve, having its motion conveyed to it by the oscillating link, is traveling at its highest velocity. This is at the exact point where the opening of the valve occurs, and coincident with the rapid movement induced by the location of the eccentric or return crank the velocity of the valve travel is further induced by the action of the combination lever, which, acting on the radius bar connection as a fulcrum, has moved the valve in its path the increased distance required to overcome the space occupied by the lap and lead. The opening of the valve, therefore, occurs more rapidly on account of the movement of the combination lever, and if we follow the movement until the middle of the piston stroke has been reached it will be found that there is a diminution of speed not only on account of the return crank passing the dead center, but this pause in the valve's motion is further accentuated by the combination lever passing the vertical dead center of its connection to the radius bar and valve rod. This double eccentricity, as has been stated, emphasizes the rapid opening and closing of the steam ports and prolongs the period of full opening. These features are of decided advantage in any kind of valve gearing used in steam engines.

In overcoming defects in construction the eccentric rod is usually the part in which any change is made. This rod may be lengthened or shortened to correct errors in the location of the main axle center or link center. The central position of the valve or the exact amount of opening of the steam port at each end of the piston stroke can readily be adjusted by lengthening or shortening the valve rod, which is usually fitted with adjustable threaded nuts. It may be added that the slightest variation of the return crank from its proper position at right angles to the main crank will be readily discovered by comparing the action of the valve as shown in the forward movement of the engine with the action and position of the valve in the backward movement. Referring again to Fig. 1, it will be readily recognized that in the event of the valve showing a greater amount of opening in the forward motion than in the backward motion the variation may be divided by moving the return crank connection the required amount nearer to the main crank, or by moving the point of suspension of the link nearer to the valve; both of these organic changes having the same effect, neither of these changes should be made except under very pressing conditions, and after repeated experiment and identical results.

Celebrated Steam Engineers.

XXI. James Nasmyth.

The adaptability of the steam engine to almost every kind of mechanical undertaking was such that it soon brought about an industrial revolution in manufacturing and transportation. In the hands of clever engineers the locomotive and steamboat soon took practical form and it is a noteworthy fact that the call for large shafting, suitable for the increasing size of steamships called into being the steam hammer. James Nasmyth, the inventor of the steam hammer, came of a Scottish family of artists, and at an early age displayed a strong bent towards engineering. He made models of engines and in 1834 he engaged in machine construction and was eminently successful as a builder of engines, boilers and machine shop tools. He was particularly successful in the manufacture of special machine tools, and was among the first to bring the turning lathe and planing machine near to that degree of perfection which these machine shop tools have subsequently attained.

It was, as we have stated, on the occasion of the necessity arising for the forging of a shaft which was to be used in a steamboat that Nasmyth's inventive mind produced the steam hammer. It immediately came into general use, and the original design is still closely adhered to. The success which attended the introduction of the steam hammer, seemed to act as an incentive to Nasmyth's constructive ability. In rapid succession he produced the steam pile-driver, thereby effecting an incalculable saving in the construction of building foundations, docks and wharves and other work, and also the flexible shaft, steam and torpedo rams, and finally the hydraulic press.

In addition to his skill as an engineer he gave much attention to the science of astronomy, and although his discoveries in connection with that science were not remarkable, he was among the first to create an interest in the phenomena known as sun spots. He was much engrossed in these speculations and many admiring friends regretted that he did not keep his fine intellect at work nearer the surface of the earth. He retired from active engineering at an early age, and devoted the latter half of his life to what may be called the pleasures of artistic and scientific speculation. It must be admitted that he did enough real work of the most enduring quality to entitle him to whatever pleasure he could find in wandering among the stars. He was born at Edinburgh in 1808, and died in 1890. He was a fine gentleman, a delightful companion, and his high intellectual qualities were enhanced by a purity of life that kept his heart young and sweet.

amount, as instances are known where crooked brake pipes containing numerous elbows have created sufficient resistance to the flow of air to prevent quick action even if the brake valve handle was placed in the emergency position.

The number of pounds difference in brake pipe and auxiliary reservoir pressures that the graduating spring is able to resist depends upon whether the triple-valve piston is moved against the graduating stem gently or with considerable force; if drawn against the stem gently, as in the case of a broken graduating pin, the spring used in the P1 or P2 (F 27 or F 29) triple valves will resist a difference of approximately 8 lbs., or if the reduction is more than 8 or 9 lbs. the spring will be compressed enough to allow the triple valve to assume quick action position. In the case of a "sticky" equalizing piston in the brake valve where the triple valves are "jerked" into application position quick action will occur with 4 or 5 lbs. reduction or almost the same instant that the equalizing piston is unseated.

DYNAMOMETER CARS.

42. A. B., St. Dominique, Que., asks: Where was the first dynamometer car built? I mean by that what road first used this kind of car? A.—The Pennsylvania Railroad was the pioneer railroad in the use of dynamometer cars. The Chicago, Burlington & Quincy followed; but their car was more or less copied from the original Pennsylvania car.

STARTING INJECTOR WHEN WATER IS LOW.

43. A. V. W., Gallitzen, Pa., writes: An engine was pushing a freight train on a 2 per cent. grade, and was stopped at a home signal. Injector was shut off and fireman discovered there was no water in the gauge and none in the water bottle. The fireman did not draw the fire, but started the injector, then went to the cabin and reported the fact to the engineer. There has been discussion among the boys of the wisdom of the fireman's act, and we wish to have your opinion. —A. The fireman did perfectly right. The difference of opinion would arise through some people holding the erroneous belief that it is dangerous to inject water upon hot sheets. The fallacy of this opinion is forcibly explained by Angus Sinclair on page 168 of "Twentieth Century Locomotives" in an article that ought to be furnished to every person connected with the operating of locomotives.

BROKEN E. T. BRAKE PIPE.

44. Subscriber, Covington, Ky., writes: When the brake pipe breaks off at the brake valve, out on the road, could not temporary repairs be made by connecting the application chamber pipe, which is copper, into the strainer where the distributing valve pipe connects, close the double cut-out cock and increase the pressure to the independent brake valve by adjusting the reducing valve and apply and release the brakes with the independent brake valve, the distributing valve being cut out?—A. The form of your question explains the method, and it is only necessary to state that you refer to the H-5 equipment, and assuming that the application chamber pipe is of sufficient length and standard or that interchangeable fittings are used. Mostly any length of train can be charged and the brake applied in this manner, and in a case of emergency, quick action could be obtained by opening the double cut-out cock and allowing the brake pipe air to escape through the broken off pipe. You will remember, of course that the H-6 equipment uses 1 in. iron pipe to convey brake pipe pressure to the distributing valve.

WHO OR WHICH?

45. Car Man, Minneapolis, writes: In several reports and circulars concerning car matters that have reached me a common expression used is, "the railroad who did the repairs." In the little grammar I studied the pronoun who was applied exclusively to persons. Inferior animals or things without life take that or which. We have been having a little dispute about the expression and decided to leave it to you. Which is correct, who or which?—A. The correct form of such a sentence as you give should use the pronoun "which."

CROSS COMPOUND PUMP CAPACITY.

46. Subscriber, Covington, Ky., writes: How would you figure the capacity of a cross compound air compressor?—A. By ascertaining the amount of free air drawn in on each stroke of the low pressure air piston. The low pressure air cylinder is 14½ ins. in diameter, the stroke is 12 ins. $14\frac{1}{2} \times 14\frac{1}{2} = 210.25$, then $210.25 \times .7854 = 165.13$, and $165.13 \times 12 = 1981.5$, and $1981.5 \times 2 = 3963$, and lastly, $3963 \div 1728 = 2.3$ cu. ft. In this, the diameter of the cylinder is squared, multiplying it by the decimal fraction .7854, which then multiplied by the length of stroke in inches gives the cubic inch capacity of the cylinder, or the amount of air in cubic inches drawn in on each stroke. The result of two strokes or one cycle, as it is called, divided by 1728, the cubic inches contained in one cubic foot, we find the pump should compres 2.3 cu. ft. of free air per cycle. The pump will actually compress 2 cu. ft. per cycle in service, which is 13 per cent. less than the theoretical value. In a future number this matter, together with some simple calculations to determine air pressures, will be referred to in a more extended manner.

BRAKE PIPE REDUCTION.

47. F. E. A., Spencer, N. C., asks: How much brake cylinder pressure should a 10-lb. brake pipe reduction put in the driver brake cylinders if the engine is equipped with the E. T. Brake?—A. 25 lbs. Every brake pipe reduction of 1 lb. should result in a brake cylinder pressure of 2½ lbs. As the application chamber is filled with atmospheric pressure before the reduction is made, there is very little additional space created by the movement of the application piston, while with a triple valve and a brake cylinder the first 10-lb. reduction is not so effective on account of the volume of air required to fill the space vacated by the movement of the brake cylinder piston.

TYPE K TRIPLE VALVE.

48. F. M., Chicago, Ill., writes: It is generally understood that brake pipe leaks have a tendency to cause undesired quick action, does not the type K triple valve in taking air from the brake pipe during service applications have a tendency to assist brake pipe leaks to throw other triples in the train into quick action?—A. No. The type K triple valve has a tendency to prevent undesired quick action by running the service application promptly through the entire train and moving all the triple valves to application position at the first reduction of brake pipe pressure. If there are enough K triple valves in the train all brakes will be applied perfectly and if brake pipe leakage reduces brake pipe pressure much faster than auxiliary reservoir pressure can expand into the brake cylinder the increasing auxiliary reservoir pressure will force the triple valve to full service position, in which the brake pipe flow to the brake cylinder is cut off and the service ports left open. This action insures the triple valves being forced to lap position when the reduction at the brake valve ceases, unless excessive brake pipe leakage will not allow them to be returned to lap position.



FRAME WELDING ON THE ERIE.

Air Brake Department

Conducted by G. W. Kiehm

Broken Air Pipes, H6 Brake

PART III.

If the brake pipe was broken or the brake valves disabled in fast passenger service while out on the road and no connection of brake cylinder pipes and brake pipe could be made between the engine and tender on account of special couplings, a connection between the main reservoir and the brake pipe could be made through the quick-action cylinder cap of the distributing valve.

It would not be a very practical method and the service application would be about as difficult as with the old style three-way cock, but the brake pipe could be kept charged and it would be an emergency brake. If the brake pipe was broken at the brake valve the brake valve would be placed on lap and the stop cock under the brake valve closed; after closing the stop cock in the supply pipe and releasing the brake, the quick-action valve and cylinder check valve can be removed, the upper cap nut removed and the application piston blocked in application position. When the adjusting nut of the safety valve is screwed down and the three caps are again screwed in place the stop cocks in the brake cylinder pipes would be closed and the stop cock in the distributing valve supply pipe opened. Main reservoir pressure will then flow through the distributing valve past the quick action and cylinder check valve seats into the brake pipe releasing the brake and recharging the train.

To apply the brake the stop cock under the brake valve would be opened and the pressure would reduce in both the brake pipe and main reservoir, unless the reservoir cut-out cock or the distributing valve cut-out cock were located within reach of the engineer or fireman; closing either cock would separate the main reservoir and brake-pipe pressures; the pump governor would regulate the pressure.

Still another connection could be made, in case of this kind, through the independent brake valve, by removing the equalizing valve from the distributing valve, closing the stop cocks in the supply and brake cylinder pipes and screwing down the adjusting nut of the reducing valve. By placing the independent brake valve in application position and adjusting the reducing valve, reservoir pressure would flow through the independent brake valve and distributing valve into the brake pipe. This could scarcely be recommended on a train of any length on account of the comparatively small ports in the distributing valve and on account of the length of time it would take to remove the equalizing valve, but if the train pipe could be charged, and it can on a short train, the brake valve stop cock can be used to assist the independent brake valve in exhausting brake-pipe pressure in cases of emergency.

If the brake pipe of H6, H5 or the standard quick-action brake is broken between the brake valve branch and the hose couplings between the engine and tender, the broken off pipe and hose can be removed and the brake pipe leak plugged, the signal and brake hose on the pilot can be coupled and the stop cock opened and the signal hose at the rear of the engine coupled with the brake-pipe hose on the front of the tender. When the brake valve is placed in release position the pressure will flow from the brake valve to the pilot, then through the signal pipe on the engine to the brake pipe on the tender and train; the brake on the engine and train can be applied in the usual manner.

This crossing over from the brake pipe to the signal pipe by hammering together the hose couplings has avoided many engine failures on account of broken brake pipes on tenders, there being but a few minutes' detention in coupling the hose.

The equalizing discharge feature of the G6 brake valve remains the same in the H5 and H6 valves; leaks about the equalizing reservoir and its connections have the same effect, and breaks in the pipe connections or reservoir itself will be handled in the same manner. Several years ago it was considered in some quarters to be desirable for the second engineer of a train double-heading to be able to note the brake-pipe pressure as maintained by the first engine, to observe the brake manipulation to a certain extent, and to apply the brake in case of emergency by a movement of the brake-valve handle alone. This was accomplished by placing the brake-valve cut-out cock in the reservoir pipe, as the brake valve was then subject to brake-pipe pressure as controlled by the first engine the black hand of the gauge would show this pressure. To prevent the equalizing piston of the brake valves from wasting brake-pipe pressure if forced from its seat during train brake release, the exhaust port was piped to a port in the cut-out cock and when the cock was closed for double-heading purposes this port would be blanked and at the same time a port at the large end of the cock would be opened, which would admit brake pipe pressure to that portion of the reservoir pipe above the brake-valve cut-out cock.

When this style of cut-out cock is used with the H6 brake the connection to the feed valve is made at a point above the cock in the reservoir pipe.

This cock is sometimes found to be leaking and the leakage is similar to that of a leakage rotary valve, leaking main reservoir pressure into the brake pipe.

The leakage if in any considerable volume sometimes lifts the equalizing piston of the brake valve.

Sometimes the cock leaks at the small end through the port through which the equalizing piston discharges the brake-pipe pressure; when it does it is a difficult matter to discover whether the leaks from the cut-out cock or from the seat of the equalizing discharge valve.

The pipes connecting the cock to the brake pipe and brake valve should be tested as well as other piping; the pipe connecting the brake pipe and the cock has pressure at all times the brake pipe is charged, the pipe connecting the brake valve and cut-out cock can be tested by closing the cock and making a service reduction with the automatic brake valve.

Should the pipe connecting the small end of the cock be broken, it would have no effect except that the air discharged by the brake valve will escape at the break instead of the port in the cock. Should the pipe connection at the large end of the cock be broken it would be necessary to stop the brake pipe leak and drive a wedge between the handle and the body of the cock to hold the plug valve to its seat.

If broken on the second engine when double-heading the brake pipe leak would be stopped and the cock handle turned to a position half way between open and closed, at which time the three ports through the cock would be closed; a wedge between the handle and the body would also be used to hold the plug valve in this position.

If at this time the brake valve rotary was forced off its seat on account of this broken pipe it would only result in a slight escape of air, or a pop at the emergency exhaust port, as the air pressure surrounding the valve would equalize instantly and seat the rotary valve before any volume could escape at the brake valve. If the pipe connecting the smaller end of the cock was broken on the second engine when double-heading, the pipe would be plugged to prevent the escape of brake-pipe pressure when the equaliz-

take, these reasons will become apparent. First, the tank may be empty or, if in the winter time, the water may become coated with a sheet of ice which will prevent the admission of air and no water can enter the suction hose. Tank hose strainer clogged with rubbish, loose hose lining, leaks in suction hose and pipe, stuck overflow valve or clogged overflow, all have a tendency to prevent the proper flow of water to the injector, and of these, dirty strainer and leaks in the suction pipe are by far the most common.

As the water begins to flow into the injector, the main steam valve is opened, the overflow closed and the injector should go to work.

In some injectors the overflow is closed mechanically in the operation of opening the steam valve or ram, while others have simply a floating valve, which closes by gravity.

The Metropolitan is a well-known example of the former and the Sellers of the latter. It is readily seen that the overflow serves a two-fold purpose: (1) It allows the air to escape from the injector when the vacuum is formed and (2) it provides for the escape of the surplus water which accumulates in the injector before the mingled steam and water has attained a sufficient velocity to overcome the opposing boiler pressure.

It is rather a remarkable fact that although the injector has been in use for over fifty years, the theory of its operation is in dispute. The "velocity" theory is the one now generally accepted as correct, although the "pressure" theory has its followers.

A significant fact in this connection is the ability of an injector to work with an open overflow. It certainly indicates that there is no pressure in the space over the tubes which would be present if the injector worked on the principle of the steam pump. After starting an injector of this type the overflow valve may be removed entirely without interrupting its working.

If an injector lifts the water and fails to force it into the boiler, its general behavior will often indicate where the trouble is. If it blows back through the suction pipe into the tank it may be due to a check valve stuck to its seat or the overflow valve may have closed or partly closed before the steam has accelerated the water sufficiently to overcome the pressure in the boiler. This seldom occurs with the open overflow, but is often very troublesome with the closed overflow; in fact, all injectors of this type in which the overflow valve and starting lever move simultaneously are objectionable for this very reason. The two valves being rigidly connected together, it is very difficult to keep a perfect adjustment and the overflow valve will sometimes seat before the main steam valve has fully opened, and then again in shutting the injector off the overflow valve may come up against its guide before the steam valve seats; result, a hot injector and a hot, sticky cab.

If part or all the water goes out of the overflow, it indicates a stuck boiler or line check, worn or obstructed tubes, leaky suction pipe, incorrect proportions of water and steam or dirt under the overflow valve. Badly worn, obstructed or loose nozzles and tubes in an open overflow injector will invariably cause a dribbling and waste of water at the overflow. The use of muddy, gritty water soon means nozzles and tubes too large or out of round, and in hard water districts they soon become limed up. This trouble increases very rapidly as the steam pressure increases.

It is important that a distinction be recognized between a faulty injector and faulty auxiliary attachments in case of failure. If the fault lies in front or behind the injector, it certainly would be a waste of time to replace it with another.

It may be said in general that if an injector stops suddenly or fails to start, the water supply has been cut off by dirt or loose hose lining. Blowing steam back through the suction pipe will generally bring relief by cleaning out the pipe and straightening out the loose lining, but it is likely to be only temporary, because the dirt will settle back again and the lining will roll up again as soon as the water begins to flow from the tank.

An injector that works intermittently and sluggishly or wastes water at the overflow generally needs repairs, although if it is getting air the same results will occur.

Injectors and accessories receive far less care and attention than they should in view of their importance. There is certainly nothing about a locomotive that could put it out of business more effectually than an interruption to its water supply. Yet there are very few injectors that are operated under ideal conditions or that do not show marks of rough usage. It would be hard to find any one thing more trying to an engineer's peace of mind than a balky injector, whether it is the fault of the injector or not. In such cases it is better policy to use the spare injector altogether before a run has to be made for water.

Right here it may be said, in passing, that the spare injector is not put on an engine as an ornament, but it should be tested frequently so that it can be relied upon to an absolute certainty in case of emergency.

The writer can recall an instance where an engineer, pulling a passenger train, had to dump his fire out in the country "miles from nowhere," for the simple reason that both injectors failed at an unexpected moment. Upon investigation he admitted that he had not tried the spare injector since receiving the engine from the shop two weeks before. When the injector was taken apart the tubes were found to be nicely coated, on the inside, with black paint which had been drawn up from the tank when the engine was tested.

In repairing injectors it is of the utmost importance to keep everything to as near standard sizes as it is possible to make it. It is an excellent policy to work from a drawing or from a standard model. Care should also be taken to use the right size of nozzles and tubes. Some injectors are made to receive nozzles and tubes of different sizes, the size determining the feeding capacity. To work economically an injector must admit of close regulation so that it may be operated



CONTRACTOR'S CAMP IN THE ROCKIES. CAN. PAC. RY.

will build up in the application chamber and apply the brake.

In this case of a broken brake pipe with the No. 6 distributing valve, the action is the same whether the plain or the quick-action cylinder cap is used, but when the quick action cap is put in use a stop cock should be placed in this branch of the brake pipe near the distributing valve, so that in case the quick-action valve was held open or the equalizing piston graduating spring was broken the brake pipe pressure could be cut off from the distributing valve. If this did occur the brake pipe pressure would be free to enter the brake cylinders and pass out the distributing valve exhaust port when the brake was released, and with no stop cock to cut off brake pipe pressure all the pressure would have to be withdrawn from the brake pipe and the branch pipe disconnected and a blind gasket inserted in the union connection; after doing this the train brakes can be released, and in making the next stop the independent brake would be used to apply the driver brake.

The slide valve in this quick action cap is liable to become leaky from dirt on its seat or being unevenly worn, and the leakage will escape past the check valve and to the atmosphere through the exhaust valve when the brake is not applied. The effect is similar to a leaky application valve, both leaking through the distributing valve exhaust port when the brake is in release and increasing the brake cylinder pressure when the brake is applied. If the volume of leakage is sufficient and the brake cylinders and pipe connections are practically free from leakage, the main piston will be moved toward release position and exhaust the increasing brake cylinder pressure to the atmosphere.

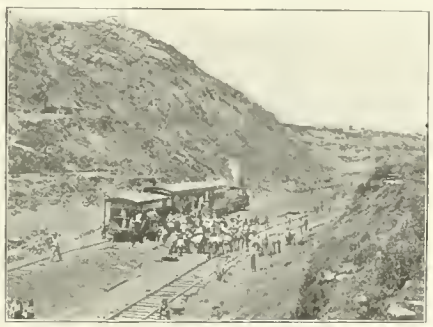
If the leakage affects the main piston in this manner the leaky valve can be detected by reducing the brake pipe pressure slightly below the adjustment of the safety valve; if the leakage then ceases it indicates a leaky quick-action valve; if the leakage at the exhaust port continues it indicates a leaky application valve.

If the leakage does not affect the main piston when the brake is applied the brake can be released and the stop cock in the supply pipe closed; after waiting a reasonable length of time for the pressure surrounding the application valve to escape, the leakage should cease if the application valve is at fault, if the leakage continues it indicates a leaky quick-action valve. During the test it should be remembered that the exhaust valve itself can leak only when there is pressure surrounding it, and at such times the brake will be applied.

The undesired quick action has been discussed to such an extent that the plain triple valve is blamed for having thrown the brakes on a train into quick action at times, therefore it is not surprising to hear that the H 5 distributing valve or the H 6 valve with the plain cylinder cap is accused of "dynamiting."

When the distributing valve is neglected until the pistons become "sticky" and hard to move, the action of the valve is sometimes similar to that of a "sticky" triple valve. The pistons of the distributing valve are sometimes allowed to get into such a condition that they cannot be moved until the gauge registers a reduction of 15 or 20 pounds, and when the piston is moved it will remain in application position until the pressure is 10 or 15 pounds stronger on the opposite side of the piston, consequently the brake will not apply promptly, and when it does apply it applies fully.

If this occurs with the automatic brake and the main piston responds promptly to the independent brake valve it indicates a sticky equalizing valve; if the condition of the equalizing valve is in question, the safety valve can be removed and the automatic brake valve used in light successive reductions, a flow of air into the passage into which the safety valve is screwed will occur the instant the equalizing valve is moved to application position.



PAY DAY IN THE CULEBRA CUT, PANAMA.

This should not be confounded with a brake that applies in full after a light reduction due to brake pipe or other leakage, such as a leaky equalizing slide valve, independent rotary valve or a leaky graduating valve in conjunction with equalizing valve packing ring leakage.

The pipes connecting the distributing valve and brake valves must be kept absolutely tight in order to maintain the brake cylinder pressure when the valve handles are on lap position; the pipes should be tested with the air pump working and excess pressure in the main reservoir; the independent brake should be applied in full, the automatic brake valve placed in driver brake holding position, and the independent valve handle returned to running position; with the H 6 brake the application cylinder pipe, the release pipe and the pipes connecting the brake valves will contain air pressure, the pump governor and feed valve maintaining the reservoir, and brake pipe pressures will prevent the equalizing valve from cutting off the application cylinder from the release pipe.

The corresponding pipes of the H 5 brake will be tested in the same manner and under the same conditions, and the double cut-out cock under the brake valve should be turned to admit air pressure to that portion of the double-heading pipe between the cut-out cock and the brake valve. Leakage in the pipes referred to or leakage past the packing leather of the application piston also has a tendency to prevent the independent brake from applying in slow application position, although this is usually found to be due to the slow application port in the independent brake valve being closed by an accumulation of dirt.

Air and Vacuum Brakes.

It is difficult to realize the amount of energy that must be destroyed by the air brake in stopping a train of cars on descending grades. The weight of the trains are often from 2,500 to 4,000 tons, and nothing but the modern air brake can be fully depended on for this work. The vacuum brake has been tested and run in competition to the air brake, but it cannot seriously compete with the former on account of the many advantages of the air brake, which deals with substance rather than void space. The air brake is more flexible; it can be applied from any portion of the train, where an opening can be made in the brake pipe; each car furnishes its own braking power, and leakage from the brake pipe has a tendency to apply the brake rather than result in a loss of train control. It can be applied instantly and released promptly on the entire train, and in stopping freight trains, the comparatively low brake pipe pressure of 70 lbs. is usually employed, and when the Westinghouse brake is used in the quick-action or emergency, position, it results in a brake cylinder pressure of approximately 60 lbs. per sq. in.; the cylinders are of 8 and 10 in., and the storage capacity of the equipments are proportioned to the size of the cylinders. The maximum pressure developed in the cylinders depends upon the amount of air that enters from the brake pipe; it is considered that the maximum pressure is about 60 lbs., whether 8 or 10-in. equipments are used, although on a train of 10 in. equipments, the cylinders would absorb more brake pipe air and have a tendency to result in a lower final equalization, the brake pipe being the same size in both cases. With the vacuum brake the higher pressures required in modern railroad service would have to be secured by increase in the size of the diaphragm and its parts.

to lap position, there must also be a leak into the ports in the valve seat to which the application cylinder and release pipes are connected.

A crack in the body gasket between the chamber and the equalizing discharge valve and the ports just referred to, would allow air to pass through the release pipe to the exhaust cavity of the brake valve in one case; in the other case, through the application cylinder pipe to the application portion of the distributing valve, thence to the release pipe to the atmosphere, and if the valve handle is moved to release, lap, or holding positions, the escape would be cut off and the brake would be applied by the building up of pressure in the application cylinder.

The leak from the rotary valve seat of the independent valve would pass into the release pipe and into the exhaust port of the automatic brake valve in one case, and into the application cylinder pipe, application cylinder, release pipe, and to the brake valve exhaust port in the other case.

The leak from the equalizing slide valve would travel the same route and would be free to pass from the pressure chamber into the application, and the supply would be maintained from the brake pipe through the feed groove, and should a leak start from any of the tubes or cored passages in the reservoir, brake pipe, pressure chamber, or supply pipe, that leakage would also escape at the automatic brake valve exhaust port and have the same effect of applying the brake when the brake valve handle is moved away from running position. Considering the leaks from the brake valve rotaries and from the equalizing slide valve as ordinary, that from the slide valve being the one usually encountered, and the possibility of leaks from the brake valve gaskets and from the reservoir passages as extraordinary, and applying the three tests, the first two of which are merely observations requiring but a moment's time, the three could be properly conducted in a very short time, admitting the possibility of the unusual being present and to avoid a complication or misunderstanding of the simple tests given, the effect of an unusual action would be considered in a measure separately from the test, but at the same time.

Suppose the blow at the brake valve exhaust port was due to a cracked brake valve body gasket which leaked, equalizing reservoir pressure into the application cylinder pipe or into the release pipe, it would be noticed during the second test when the brake valve handle was moved to lap position, by the equalizing discharge valve becoming unbalanced, due to the reduction of equalizing reservoir pressure, and the result would be a discharge of air at the brake pipe exhaust port, and an automatic application of the brake. Should the supply pipe passage of the distributing valve reservoir start a leak into the application chamber, it would upset the theory advanced in the first observation or test, and indicate a leaky automatic brake valve rotary, but during the second test when the independent brake valve is placed on lap position to separate the automatic brake valve and the distributing valve, it would be observed that the application cylinder pressure was built up to the adjustment of the safety valve by this leak of main reservoir pressure, which we know could not occur from an equalizing slide valve leak under a 70-lb. brake pipe pressure.

The effect of a leaky brake pipe passage and a leak in the pressure chamber tube where they pass through the application chamber, would be similar to a leaky equalizing slide valve, and in the first test, just to what pressure the brake cylinder hand on the gauge would rise to, depends upon the amount of leakage and the length and size of the brake pipe, for if the volume of leakage from the equalizing slide valve is about equal to the capacity of the feed groove the pressure chamber could not be charged, and a 10-lb. reduction would not move the equalizing valve, but the brake would of course be applied by the leakage, and in a short time the brake pipe, pressure chamber, and application cylinder pressures would equalize.

If a leak in the tube or cored passage should exist, according to the test, it would indicate a leaky equalizing slide valve, and if the distributing valve were removed and replaced by one tested and known to be in good condition and the same disorder continued and under the same conditions, a test for the tube and passage would be necessary to locate the defective one, or a test would be necessary to locate a leak or crack that would allow pressure from an unusual source to enter the application chamber. The reservoir can be thoroughly tested without removing it from the engine by whittling a few wooden plugs to fit the different sized ports exposed after the distributing valve has been removed from the reservoir, and before removing it the brake would be applied in full by exhausing the brake pipe pressure; then the stop cocks in the reservoir supply pipe and brake cylinder pipe can be closed and the distributing valve removed, driving plugs in each port successively or bolting a circular board with an opening in the center, and a blind gasket to cover the various ports, to the distributing valve reservoir and opening first the stop cock in the brake cylinder pipe will allow brake cylinder pressure to flow back to the cored passage in the reservoir, the reservoir or supply passage can then be tested by opening the cock in that pipe, placing the brake valve handle in running position would charge the brake pipe passage.

After those parts are known to be free from cracks or flaws, the pressure chamber tube and the wall dividing the application and pressure chambers can be tesed by making a connection from the signal line to the pressure chamber drain passage, by means of a hose, a piece of pipe, and a few fittings.

Of course, it is understood that a leak of this kind might never occur in the entire experience of an individual, but is liable to occur at any time, and to thoroughly understand the effects of any leaks that are liable to occur will lead to a pretty fair knowledge of the construction and operation of the air brake, and when that is once understood the so-called mysterious actions of the brake will disappear.

The leak through the brake valve gasket is not so unusual, and is liable to be met with at any time, and the leaks from the rotary valve seats of the brake valves are the natural results of wear, while the leak from the equalizing slide valve of the distributing valve is of the most common occurrence, being similar to a triple valve, will leak from the same causes that triple slide valves will, and if the action is always compared with that of a triple valve on a car, and the brake valves with that of a retaining valve, and the application piston with the brake cylinder piston, there will be no difficulty encountered in locating the cause of an unusual action. Unfortunately for those who do not follow air brake matters closely, the H6 and the H5 brake equipments are in use at the same time, and often both are found on the same class of locomotives, however the different parts of the equipments are not interchangeable, and the differences are not so marked but that a knowledge of one will quickly lead to an understanding of the other, it being remembered that in the H5 brake the application cylinder and the application chamber are always in communication and open to the atmosphere when the brake valve handles are in their running positions, while in the H6 brake those chambers are open to the atmosphere through the exhaust port of the equalizing valve when it is in release position, both valve handles being in running position, and the application cylinder and pressure chamber are divided when the equalizing valve is in emergency position. In the H6 distributing valve there is no port opening in its emergency position to regulate application cylinder pressure, but instead a "blow down timing port" is added in the brake valve, it and the warning port receiving their supply of air from the feed valve pipe. The number 6 distributing valve reservoir has a 1-in. brake pipe connection to provide a large brake pipe opening for emergency applications when the quick-action cap is used, in double heading the stop cock under brake valve is closed, but the valve handle remains in running position.